DEAL PARTNER & ENGINEERING REMEDIATION MEMORANDUM

CDD Deck Gap Analysis ("For Dummies") & Engineering Blueprint

Forensic Comparison of Benchmark Analyst Deck (Compost_Crew_CDD_v4_2.pptx) vs. Automated Pipeline Output (Test6_CDD_Deck (1).pptx)

TARGET ENTITY	BENCHMARK DECK (V4_2)	AUTOMATED DECK (TEST6)	CORE DIAGNOSIS
Compost Crew Inc. (Series B)	**55 Slides** · 9.2 / 10 (IC Ready)	**29 Slides** · 5.2 / 10 (Starved)	**26 Slides Dropped by Split-Brain Bug**

1. Executive Summary ("For Dummies"): Why Are the Two Presentations So Different?

Think of our automated system as two teams: **35 Junior AI Researchers (the Agent Modules)** who read the 40-file data room and fill out worksheets, and a **Slide Designer (the Test6 Deck Compiler)** that turns those worksheets into a PowerPoint presentation for the Investment Committee (IC). Compared to the broken v1 deck, **Test6 fixed the slide order and got the basic P&L and TAM numbers right**, but still lost 26 slides of vital commercial analysis for four simple reasons:

BOTTLENECK 1 · RC-1

The "Bottom-of-the-Page" Bug

Our AI researchers actually found the right answers in the data room, but wrote them in **paragraphs at the bottom** of their worksheets while leaving the **top summary tables blank**. Because the top tables were blank, the worksheets stamped themselves `BLOCKED`.

BOTTLENECK 2 · RC-1

26 Slides Thrown in the Trash

The **Test6 Slide Designer** only reads the top tables of the worksheets. When it saw `BLOCKED` on 8 major modules (Customers, Competitors, Growth, Pricing, Risks), it obeyed its safety rules and **deleted 26 slides** rather than guessing.

BOTTLENECK 3 · RC-2

Only 1 of 12 Excel Tabs Read

The company's main Excel file (`CC_Investor_Workbook`) has **12+ tabs** (covering DC contract scenarios, sales quotas, pricing, and customer cohorts). Test6 only read the main P&L tab and ignored the other 10 tabs.

BOTTLENECK 4 · RC-3

No "Calculator" Reality Check

Test6 passively copies management's numbers onto slides (*"Plan is $50.2M"*). Benchmark **v4_2** pulls out a calculator and proves management's math doesn't add up (*"At 3% YTD growth, you only hit $15.4M, and 81 new logos equals just 1.6 sales reps!"*).

2. Table 1: Side-by-Side Comparison, Plain-English Gaps & Root Causes (v4_2 vs. Test6)

REPORT SECTION	WHAT BENCHMARK V4_2 SHOWS (55 SLIDES — RATED 9.2 / 10)	WHAT AUTOMATED TEST6 SHOWS (29 SLIDES — RATED 5.2 / 10)	THE GAP (IN PLAIN ENGLISH)	ROOT CAUSE ("WHY IT BROKE")
Overall Scorecard	**Complete IC-ready book.** Answers every question on financials, customers, competitors, risks, and the 2030 plan.	**Half-empty skeleton.** Clean layout and accurate top-line P&L, but missing 65% of the commercial story.	**26 slides of core commercial analysis are missing** in Test6.	RC-1: Split-Brain Bug Agents put answers in bottom prose, left top tables blank, and triggered `BLOCKED` gates.

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01. Business & Cost Structure	**5 slides:** P&L, company milestones, exact 2025 cost structure (Own crews 33.3%, Subcontracted 8.8% [$1.18M], Fleet 9.4%, Processing 9.7%, SG&A 31.5%), and segment unit economics ($566/mo commercial vs. $29.19/mo residential).	**1 slide:** Only shows top-line revenue ($7.6M, $11.9M, $13.4M) and prints raw database log strings like `Databook promoted · 28 P&L line(s)` on Slide 8.	Test6 never tells the IC what the company spends money on, how many trucks it runs (40), or that 64% of revenue has **no contract lock-in** (pay-as-you-go).	**RC-2: Workbook Blindness** **RC-4: Telemetry Leak** Ignored COGS & Unit Economics tabs; leaked internal log text onto Slide 8.
02. Investment Highlights	**4 slides:** 6-row executive scorecard, YoY growth decomposition by segment (+$4.31M in '24 vs. +$1.54M in '25), and a 7-row fact-check of CIM claims (flagging 99.5% on-time & 70+ NPS as unverified).	**2 slides:** Shows headline TAM/SAM/SOM ($1,546M / $464M / $139M) and total revenue growth (+$4.31M, +$1.54M) with zero breakdown by customer segment.	Test6 fails to tell the IC that **Residential stopped growing** (+$0.01M in 2025) or that 2024's 57% jump was a one-time step-up from the DC contract.	**RC-2: Workbook Blindness** Did not pull the `Revenue by Segment` tab into the YoY growth slide.
03. Market Size & 2030P Plan Check	**7 slides:** Critiques the TAM model (notes **Municipal—35.7% of revenue—is excluded**), breaks down the $50.2M bridge, and proves that 2025/2026 momentum only yields **$15.4M–$24.5M** and DC bills $3.18M/yr vs. $4.0M plan.	**4 slides:** Copies TAM/SAM/SOM and subtracts 2025 QBO revenue ($13.41M) from 2030P Model revenue ($50.17M) to get +$36.76M (3.74x)—with **zero driver breakdown and zero momentum check**.	Test6 accepts management's $50.2M forecast uncritically and mixes QBO actuals ($13.41M) with Model basis ($13.28M) in the subtraction.	**RC-3: No Calculator Node** Copies numbers passively without running cross-check math (momentum extrapolation & basis checks).
03. Commercial Risk Register	**1 full slide (9 risks):** Quantifies and scores top deal killers (`Plan execution: 20`, `DC concentration: 15`, `DC durability: 15`, `Commercial logo drop: 12`, `Thin earnings: 12`).	**1 broken slide (Slide 16):** Prints an empty table whose only row literally reads: `Market Risk for Co. 0 external risk(s). Internal prior=yes`.	Test6 tells the IC there are **zero external risks** in the deal and prints a raw software debug message as the risk name.	**RC-1: Split-Brain Bug** **RC-4: Telemetry Leak** `Market Risk` top table was blank; renderer printed the agent's status header.
04. Competitors & Pricing	**4 slides:** Maps 4 real local DMV rivals (**Compost Cab, Veteran Compost, Monster Organics, Pluvr**), critiques CIM p.53 self-assessment, benchmarks Residential ARPU ($29.19 vs. Compost Cab's $32/mo), and uncovers **2,071 cheap ($1.94/mo) municipal homes** in Stripe.	**1 slide (Slide 18):** Shows a 1-row table of Commercial retention (GRR 87.3%, NRR 106.6%, Spread +19.3 pts). **Zero competitors, zero pricing slides, and zero residential retention**.	Test6 has **no competitive landscape and no pricing benchmark at all**.	**RC-5: No Local Web Fallback** **RC-1: Split-Brain Bug** VDR had no competitor pack; v4_2 used geo-fenced DMV web research, whereas Test6 just blocked the section.
05. Growth Levers, Sales Team & M&A	**8 slides:** 8-lever 2030P bridge (+$36.89M); proves 2025 commercial wins (81 logos) equaled **just 1.6 reps' quota** (while active accounts fell 663 to 591); GTM rep ramp (8 to 23 reps); Top 9 MD/NOVA ZIPs (`22201`); Utica ($1.64M) & Agricity M&A split ($0.59M bought vs. $0.91M gross synergy).	**2 slides (Slides 20–21):** 1 sentence listing commercial net additions (139, 95, 136, 33) and 1 commercial retention chart. **Omits all 6 other growth slides**.	Test6 omits the 8-lever revenue bridge, sales rep & quota math, pricing escalators, ZIP code rankings, Utica buildout, and Agricity M&A analysis.	**RC-1: Split-Brain Bug** **RC-2: Workbook Blindness** `Growth Opportunities` marked itself `BLOCKED`; ZIP workbook & CIM p.29 were not promoted to Databook.

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07. Appendix B (Audit Trail)	**9 slides:** Module coverage scorecard, 40-file VDR source register, and a 19-row graded Exceptions Register (`CRITICAL / HIGH / MEDIUM`) explaining every conflict resolved.	**3 slides (Slides 27–29):** Cleanly placed at the end of the deck, but simply lists 8 modules as `Blocked — Findings from it are not relied on`.	Test6 gives up on 8 blocked modules instead of pulling fallback facts directly from the VDR workbooks like v4_2 did.	RC-1: Split-Brain Bug No fallback rule to query the Databook when an upstream agent table is empty.

3. Table 2: Engineering Action Plan — How to Fix Every Root Cause

ROOT CAUSE	WHAT IS BREAKING IN THE CODE	AFFECTED PIPELINE COMPONENT	STEP-BY-STEP IMPLEMENTATION INSTRUCTIONS FOR ENGINEERS
RC-1 **Split-Brain Agent Serializer**	Bottom RAG prose extracts VDR facts, but top Markdown/JSON tables run independently and stay blank (`0 rows`) → triggers `Reliance: BLOCKED` → Test6 deletes 26 slides.	Agent Runner Orchestrator (`customer_segmentation`, `stickiness`, `growth_opportunities`, `competitive_diff`, `market_risk`, `demand_drivers`)	**1. Invert Execution Graph:** Run the RAG sub-queries *first*. **2. Add Pydantic Table Mapper:** Fee RAG prose answers + VDR chunks in strict structured-output LLM call (`response_format={"type": "json_object"}`) that populates the schema tables (Sections 1–6). **3. Delay QA Gate Evaluation:** Comp `Quality` and `Reliance` (`READY / LIMITED / BLOCKED`) *only after* the tables are populated from the RAG p strip the redundant raw Q&A dump fr bottom of the `.md` file.

ROOT CAUSE	WHAT IS BREAKING IN THE CODE	AFFECTED PIPELINE COMPONENT	STEP-BY-STEP IMPLEMENTATION INSTRUCTIONS FOR ENGINEERS
RC-2 **Single-Tab Workbook Blindness**	Databook promoter only promotes the summary P&L tab and ignores 10+ commercial, cohort, scenario, and ZIP tabs in the VDR Excel workbooks.	Databook Ingestion Engine (`compost-crew-databook` / `openpyxl` parser)	**1. Promote All Investor Workbook** Write deterministic `openpyxl` parse promote all tabs of `CC_Investor_Workbook_EXTERNAL` (`Revenue by Segment` , `Unit Eco` `Commercial Summary` , `Commercia` `Concentration` , `Residential Su` `DC 2030 Scenarios` , `CIM Crossw` and `MD_NOVA_ZIP Expansion Analysis.xlsx` . **2. Promote QBO COGS Lines:** Extra value-chain cost lines from `Compost` `Monthly Financials 23YTD26.xl` **3. Wire Databook Fallback:** If an ag module returns `<40%` coverage, the compiler must pull directly from these promoted Databook tables.

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RC-3 **No Forensic Calculator Node**	Test6 only copies numbers from single tables; it never runs cross-document math checks or verifies QBO-vs-Model basis consistency.	Post-Databook / Pre-Deck Analytical Synthesis Layer	Build a deterministic Python **Forensi Check Node** that computes 5 standa checks: **1. Basis Lock:** Subtract 2025 *Model* ($13.28M) from 2030P *Model* Revenu ($50.17M) = `+$36.89M (3.78x)`; n QBO ($13.41M) and Model bases in bridge. **2. Momentum Extrapolation:** Comp `Rev_2025 × (1 + g_2025)^5` ($2 and `Rev_2025 × (1 + g_YTD26)^` ($15.40M) vs. 2030P Plan ($50.17M) **3. Run-Rate vs. Plan Check:** Annua DC billings (`$1.59M × 2 = $3.18M` $4.00M 2026P target. **4. Sales Quota Check:** Divide 2025 commercial logos (`81`) by AE annua (`52`) = `1.56 reps' quota`. **5. Population Reconcile:** Diff Stripe subscriptions (`7,566`) vs. paying res book (`5,495`) to flag the `2,071` nor rate ($1.94/mo) municipal homes.

ROOT CAUSE	WHAT IS BREAKING IN THE CODE	AFFECTED PIPELINE COMPONENT	STEP-BY-STEP IMPLEMENTATION INSTRUCTIONS FOR ENGINEERS
RC-4 **Raw Pipeline Telemetry Leaks**	Slide renderer binds internal `insight_snapshot` status strings (`0 external risk(s). Internal prior=yes` , `Databook promoted`) directly into slide bullets and risk tables.	PPTX Slide Renderer & Risk Register Aggregator	**1. Decouple Telemetry from Slide T** Never map the internal `insight_sn` diagnostic string into slide body bulle table cells. **2. Add Regex Sanitizer Gate:** Block string matching `Internal prior=` , `Databook promoted` , `accounting period(s) published` , `QA \d/\d` or `Information request:` from re on content slides. **3. Cross-Module Risk Aggregator:** the Section 03 Risk Register slide by aggregating scored risks (`Impact 1 Likelihood 1–5`) across all workstr (Plan execution, DC concentration, Commercial logo drop, Thin EBITDA) than reading a single empty `Market` table.
RC-5 **Missing Geo-Fenced Web Fallback**	When VDR contains no third-party competitor pack, `Competitor Identification` and `Market Pricing` block themselves, leaving Section 04 empty.	`competitor_identification` & `market_pricing` Agents	**1. Conditional Geo-Fenced Search** VDR named competitor count == `0` , automatically trigger a web search str constrained to the verified `Market Definition` geography (`"Washing OR "Maryland" OR "Northern Virginia"`) + service (`"compost" "food waste collection"`). **2. Entity De-Confliction Filter:** Excl target-owned brands (`KeyComposta` past acquisitions (`Fat Worm` , `Key C` and subcontractors (`Agricity`) fron competitor list. **3. Scrape Published Local Prices:** published residential/commercial list ((e.g., `Compost Cab $32/mo`) and p `Market Pricing` graded as `indic (web, dated)` .